DE HAVILLAND AIRCRAFT OF CANADA LIMITED

# TEMPORARY REVISION

**PSM: 1-64-2** **AIRCRAFT: DHC-6** **SERIES: 400**

**PUBLICATION: MAINTENANCE MANUAL** **TEMP. REVISION NO.: TR76-4**

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Insert in 76-10-00 before Page 1

POWER CONTROL

**REASON FOR ISSUE:**

To introduce S.O.O. 6443, Introduction of Optional Power Lever Rigging Procedure.

**ACTION:**

In Chapter 76, Section 76-10-00, ignore existing pages 1 and 2 and insert revised pages as shown on Pages 2 to 4 of this Temporary Revision.

**-Continued overleaf-**

**RECORDING INSTRUCTIONS:**

Record this TEMPORARY REVISION NO. and the relevant information in the appropriate columns of the TEMPORARY REVISION INDEX provided at the front of the manual.

Power Control – Maintenance Practices

1. General

Control of an engine is achieved through the operation of the power lever control system, the fuel shut-off control system and the propeller control system. The control levers for each system are mounted in the flight compartment overhead console. A cable attached to each lever quadrant runs along the aircraft roof, wing leading edge, and engine nacelle to its respective control pulley on the engine firewall. From the control pulleys, a rod or teleflex is used to actuate the fuel control unit, fuel shut-off valve and the propeller governor. Both engines are controlled by identical systems.

**CAUTION** **DO NOT SELECT REVERSE POWER WITH ENGINE STOPPED, IF BETA CONTROL IS CONNECTED.**

2. Rig Power Lever Controls

Refer to Figure 1 for details.

Note Slotted adjustment points utilize serrated washers. Ensure serrations in vernier washer and lever are properly aligned before tightening bolts.

A. Adjust Linkage and Controls at Rear of Engine

Adjust the linkage and controls at the rear of the engine as follows:

1) Disconnect the rear teleflex control rod at the beta control cam. If the engine is installed, ensure that the power pushrod is disconnected at the cam follower lever.

2) Disconnect the FCU control rod from the FCU arm, on the FCU assembly.

3) Remove the nut, washer and cotter pin from the FCU spindle shaft.

4) Install rigging gauge SD12556 so that the maximum power stop bolt is hard against the rigging face and the gauge bush screws onto the FCU spindle shaft, but allowing the serrated vernier washer and the FCU arm to rotate freely on the spindle.

Note Ensure that the vernier washer is positioned with the 25–serration side toward the FCU arm.

5) Position the FCU arm behind the hole in the rigging gauge; insert the gauge rigging pin through the hole and FCU arm.

6) With the maximum power stop bolt hard against the rigging gauge face, rotate the vernier washer until correct meshing with the FCU arm and spindle is obtained. Mark the relative position between the FCU arm, vernier washer and FCU spindle with permanent marker.

7) Adjust the low-idle set screw to contact the FCU stop lever, and lock it in position.

8) Remove the rigging gauge. Ensure that the low-idle set screw contacts the FCU stop lever and the alignment marking obtained in Para 2.A.6) is in line. Install the castilated nut and cotter pin on the FCU spindle shaft.

9) Connect the FCU control rod to the midslot position on the FCU arm.

10) With the low-idle set screw contacting FCU stop lever and cam follower pin contacting the FCU actuating lever, adjusting the length of the FCU control rod until the index marks are in line, between cam follower pin and FCU actuating lever.

11) For aircraft with the basic reverse pickup configuration:

With the FCU reversing lever contacting the base of the FCU actuating lever and the cam index marks in line, set the stop bolt on the reversing lever to give a 0.35 inch gap between the bolt head and follower pin.

For aircraft configured with delayed reverse pickup (S.O.O. 6443 required):

With the FCU reversing lever contacting the base of the FCU actuating lever and the cam index marks in line, set the stop bolt on the reversing lever to give a 0.44 inch gap between the bolt head and follower pin.

Note Both engines must be rigged in the same (basic or delayed) reverse pickup configuration, asymmetric rigging is not allowed. Aircraft with S.O.O. 6443 installed may be rigged to the basic reverse pickup configuration if both engines are rigged as such.

12) Verify that marked position Para 2.A.6) has remained in alignment, if not repeat procedures from Para 2.A.6) through Para 2.A.12).

13) Reinstall rear teleflex control rod to beta control cam and secure all applicable hardware.

14) Ensure all applicable hardware is secured.

Note Beta Controls to be reconnected and adjusted as necessary to Rig Beta Controls System (refer to Para 3.A).

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9) Connect the FCU control rod to the midslot position on the FCU arm.

10) With the low-idle set screw contacting FCU stop lever and cam follower pin contacting the FCU actuating lever, adjusting the length of the FCU control rod until the index marks are in line, between cam follower pin and FCU actuating lever.

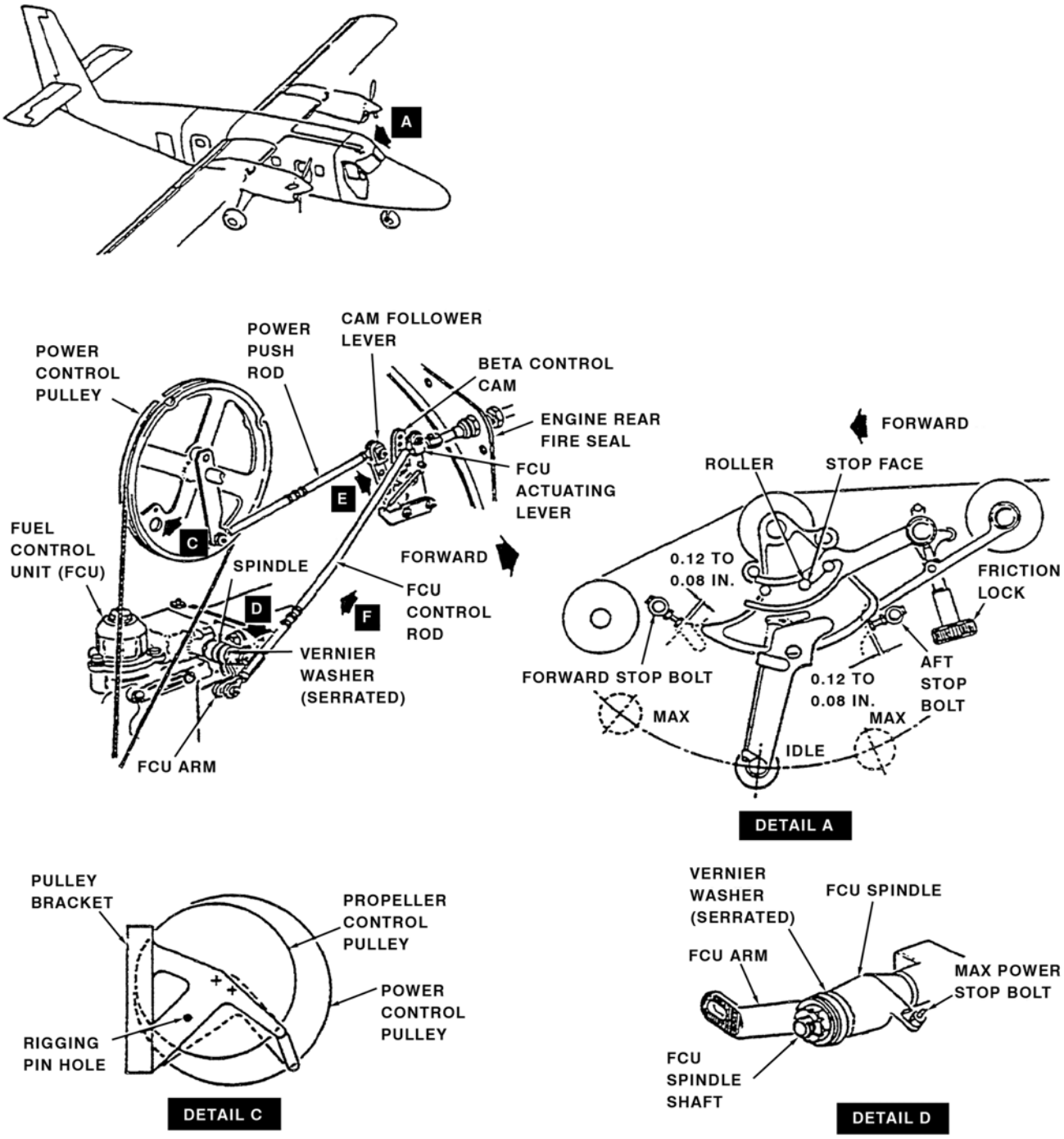
11) With the FCU reversing lever contacting the base of the FCU actuating lever and the cam index marks in line, set the stop bolt on the reversing lever to give a 0.35 inch gap between the bolt head and follower pin.

12) Verify that marked position Para 2.A.6) has remained in alignment, if not repeat procedures from Para 2.A.6) through Para 2.A.12).

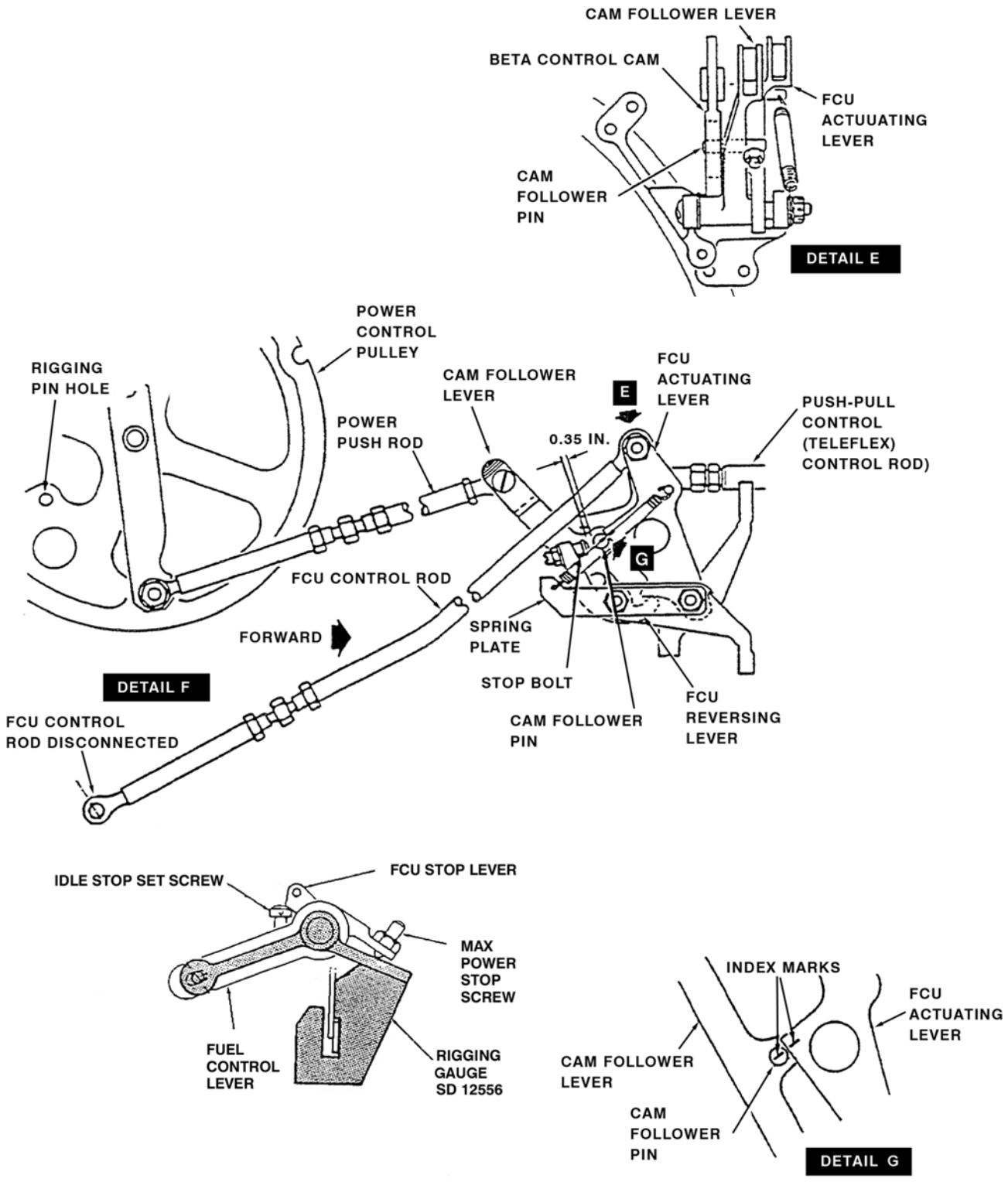
13) Reinstall rear teleflex control rod to beta control cam and secure all applicable hardware.

14) Ensure all applicable hardware is secured.

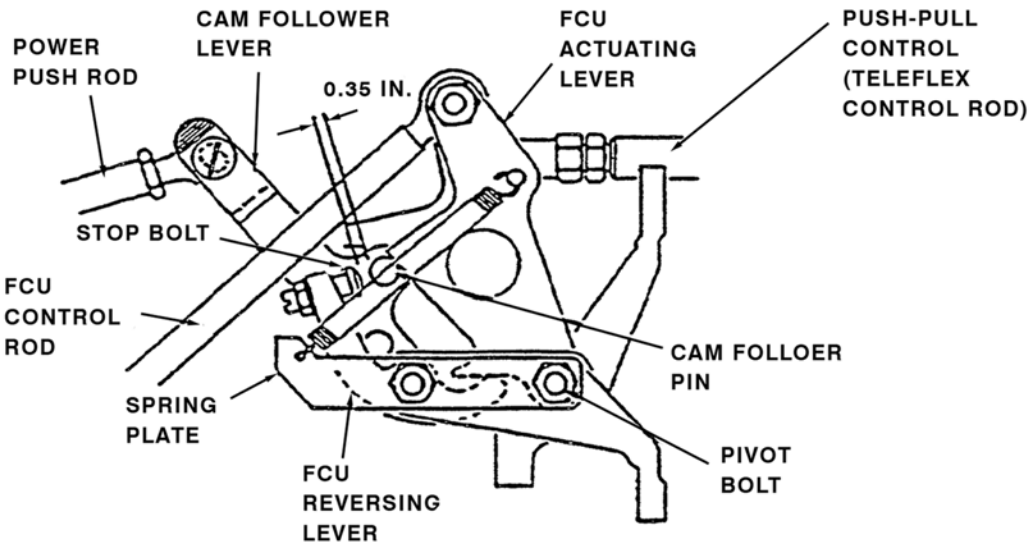
Note Beta Controls to be reconnected and adjusted as necessary to Rig Beta Controls System (refer to Para 3.A).




Power Lever Control System Rigging
Figure 1 (Sheet 1 of 3)




Power Lever Control System Rigging
Figure 1 (Sheet 2 of 3)




Power Lever Control System Rigging
Figure 1 (Sheet 3 of 3)

B. Adjust Controls in Airframe

With the power push rod disconnected at cam follower lever.

1) Move power lever in flight compartment to IDLE position (roller against stop face). Apply friction lock.

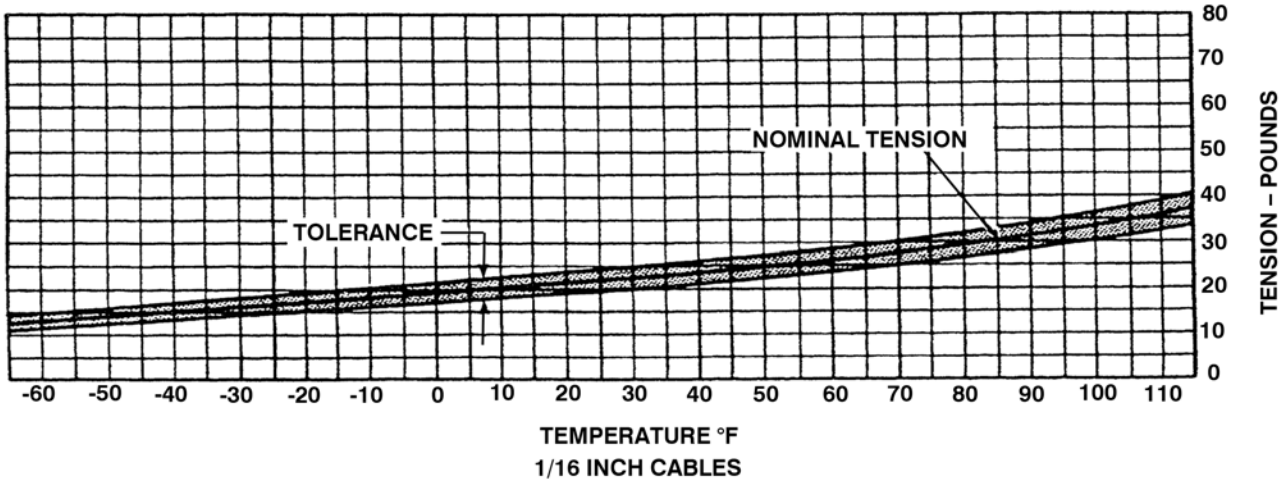
2) Insert rigging pin SD10545–1 through propeller control pulley and into power control pulley in the engine nacelle.

3) Adjust turnbuckles in wing leading edge to obtain cable tension shown in Figure 2.

4) Ensure power lever is not moved from IDLE and rigging pin moves freely in and out of the rigging pin hole of power control pulley.

5) Lock turnbuckles. Remove rigging pin.

6) Operate lever through full range of travel, return to IDLE and ensure that rigging pin can be freely inserted into rigging hole. Readjust turnbuckles if necessary.




Cable Tension Graph
Figure 2

C. Adjust Controls – Airframe to Engine Interconnect

1) With linkage and controls rigged as detailed in Para 2.A, position FCU stop lever to contact low idle set screw.

Note Ensure the rear teleflex cable rod end is disconnected at the beta control cam.

2) With rigging pin SD10545–1 installed in power control pulley and cam follower pin contacting FCU actuating lever, adjust length of power push rod (if necessary) and connect to mid-slot of cam follower lever.

Note Cam follower pin index mark and FCU actuating lever index mark should remain aligned from Para 2.A.10).

3) Remove rigging pin from power control pulley.

4) Advance power lever in flight compartment until max power stop bolt contacts maximum power stop anvil on the fuel control unit.

5) Adjust and lock power lever forward stop bolt in overhead console to give 0.12 to 0.08 inch clearance between stop bolt and power lever (refer to Figure 1, Detail A). Clearance may be checked with setting pin SD12570–3 or feeler gauge.

6) Set propeller lever to full INCREASE and retard power lever to reverse in the flight compartment until max power stop bolt again contacts maximum power stop anvil.

7) Adjust and lock power lever aft stop bolt in overhead console to give 0.12 to 0.08 inch clearance between stop bolt and power lever. Clearance may be checked with setting pin SD12570–3 or feeler gauge.

8) Move power lever through full range of travel and check for complete freedom of movement.

9) Connect rear teleflex push-pull control rod end to beta control cam.

Note On completion of adjustment, a check should be carried out to ensure that FCU actuating lever and cam cluster geometry arrangement look the same on both engines. Any appreciable difference will induce power lever split.

10) Rig Beta controls in accordance with Para 3.A (Pre Mod 6/2476) or Para 3.B (Post Mod 6/2476).

Note Refer to 76-10-11, Overhead Console – General Data for rigging of power/propeller lever interlock, and to Para 6., for rigging of propeller reset and autofeather microswitches.

3. Rig Beta Controls

A. Adjust Engine Front and Rear Linkage (Pre Mod 6/2476)

Refer to Figure 5 for details.

1) Disconnect teleflex push-pull control rod end at beta control cam, at rear of engine.

2) Disconnect fuel governor air bleed link at fuel governor reset arm.

3) Adjust propeller governor beta valve:

a) Loosen low pitch adjuster locknut.

b) With propeller in FEATHER and while holding a forward force on propeller reversing lever, adjust low pitch stop in or out, as necessary, to set the bottom of beta valve clevis slot flush with front face of cap nut.

c) Tighten low pitch adjuster locknut.

d) Disconnect push-pull rod from propeller reversing lever.

Note Ensure bushing is not lost from reversing lever when disconnecting push-pull rod.

e) Check that 1.1 to 1.3 inch travel exists in low pitch adjuster by sliding push-pull rod in and out of adjuster.

f) On satisfactory completion of check, re-assemble control at propeller reversing lever.

Note If 1.1 to 1.3 inch travel cannot be obtained check position of propeller low stop collar; distance from propeller mounting face to forward face of carbon block groove should be 1.26 to 1.27 inches. Adjust low stop rods (three) in collar as necessary (refer to Hartzell Manual).

Wear of low stop collar and of cam follower pin in beta control cam will give a very low $N_P$ at 52% $N_G$.

4) With linkage still positioned as in Para 3.A.3), move fuel governor reset arm forward against its maximum stop. Adjust length of fuel governor link and connect to inner hole of fuel governor reset arm.

5) Move power lever in overhead console fully forward and apply friction lock.

6) Ensure reversing lever is fully forward and low pitch stop in adjuster is contacted and fuel governor reset arm is set as per (Para 3.A.3)). Adjust length of push-pull control at rear clevis rod end to align with the hole in the beta control cam, then lengthen push-pull control by 1/2-turn of rear clevis rod end to ensure low pitch stop is contacted, and connect to hole in beta control cam.

7) Release power lever friction lock, move power lever through full forward range of travel and check for binding and complete freedom of movement. Do not operate in reverse range.

8) Move power levers of both engines back to idle. Check that push-pull teleflex control in low pitch stop adjuster of both engines move back the same distance (approx. 1/8). If required, adjust push-pull at rear clevis rod end to obtain similar movement on both engines.

B. Adjust Engine Front and Rear Linkage (Post Mod 6/2476)

Refer to Figure 6 for details.

1) Disconnect teleflex push-pull control rod end at beta control cam, at rear of engine.

2) Disconnect fuel governor interconnect rod at fuel governor reset arm.

3) Adjust propeller governor beta valve:

a) Loosen low pitch adjuster locknut.

b) With propeller in FEATHER and while holding a forward force on propeller reversing lever, adjust low pitch stop in or out, as necessary, to set the bottom of beta valve clevis slot flush with front face of cap nut.

c) Tighten low pitch adjuster locknut.

d) Disconnect push-pull rod from propeller reversing lever.

Note Ensure bushing is not lost from reversing lever when disconnecting push-pull rod.

e) Check that 1.1 to 1.3 inch travel exists in low pitch adjuster by sliding push-pull rod in and out of adjuster.

f) On satisfactory completion of check, re-assemble control at propeller reversing lever.

Note If 1.1 to 1.3 inch travel cannot be obtained check position of propeller low stop collar; distance from propeller mounting face to forward face of carbon block groove should be 1.26 to 1.27 inches. Adjust low stop rods (three) in collar as necessary (refer to Hartzell Manual).

Wear of low stop collar and of cam follower pin in beta control cam will give a very low $N_P$ at 52% $N_G$.

4) Adjust fuel governor interconnect rod:

a) Separate the fuel governor interconnect rod, and install the governor rod end and internal adjuster to the inner hole of the fuel governor reset arm. Secure with bolt, washer, and castellated nut. Tighten the castellated nut and torque to 12 to 18 in–lbs. Lock with a cotter pin. Refer to Detail C of Figure 6.

b) Re-connect the two halves of the fuel governor interconnect rod. With the interconnect rod still positioned fully forward and the beta valve slot flush with the front face of the cap nut, move fuel governor reset arm fully forward against its maximum stop.

c) Adjust the fuel governor interconnect rod to set a 0.100 inch ± 0.010 inch gap between the internal and external adjusters. Refer to Section D–D of Figure 6.

d) Check upper rod end for safety. Separate the fuel governor interconnect rod to verify 0.300 inch minimum engagement of internal adjuster at lower rod end. Refer to Section D–D of Figure 6.

e) Tighten locknuts 32 to 36 in–lbs and fasten with lockwire.

Note For fuel governor interconnect rod lubrication instructions, refer to Pratt & Whitney PT6A-34 Maintenance Manual.

5) Move power lever in overhead console fully forward and apply friction lock.

6) Ensure reversing lever is fully forward and low pitch stop in adjuster is contacted and fuel governor reset arm is set as per (Para 3.B.3)). Adjust length of push-pull control at rear clevis rod end to align with the hole in the beta control cam, then lengthen push-pull control by 1/2-turn of rear clevis rod end to ensure low pitch stop is contacted, and connect to hole in beta control cam.

7) With the power levers at idle, check the fuel governor interconnect rod to ensure the gap between the internal and external adjusters is 0.000–0.003 inch, and ensure fuel governor reset arm is against the reset stop. Refer to Detail C of Figure 6.

8) Release power lever friction lock, move power lever through full forward range of travel and check for binding and complete freedom of movement. Do not operate in reverse range.

9) Move power levers of both engines back to idle. Check that push-pull teleflex control in low pitch stop adjuster of both engines move back the same distance (approx. 1/8). If required, adjust push-pull at rear clevis rod end to obtain similar movement on both engines.

4. Rig Fuel Shut-off Controls

Refer to Figure 3 for details.

A. Adjustment of the Fuel Shut-off Controls in Airframe

1) Disconnect the FCU fuel shut-off rod end at the intermediate lever attachment bolt on the FCU unit.

2) Select the fuel condition lever to the OFF position in the flight compartment.

3) Install the rig pin in the nacelle engine pulley quadrant.

4) Adjust the wing leading edge control cable and turnbuckles for correct cable tension as required (approximately 25 lbs), refer to Figure 2.

Note Ensure that the fuel condition lever remains in the OFF detent during this adjustment.

5) Remove the rig pin in the nacelle.

6) Select the fuel condition lever to the ON position.

7) Ensure engine control lever moves freely with no binding.

8) Safety the engine control cable turnbuckles.

B. Adjustment of the Fuel Shut-off Control Airframe to Engine

1) Ensure the engine fuel control lever and fuel shut-off control system are rigged according to Para 4.A.

2) Move the engine fuel condition lever in flight compartment to the OFF position.

3) Install the rig pin in the nacelle engine pulley quadrant.

4) Position the FCU intermediate lever to maintain contact between the FCU actuating lever and the FCU fuel cut-off stop bolt.

5) In the nacelle, adjust the fuel shut-off control rod to allow the attach bolt to be installed while maintaining the contact between the FCU actuating lever and the FCU fuel cut-off stop bolt.

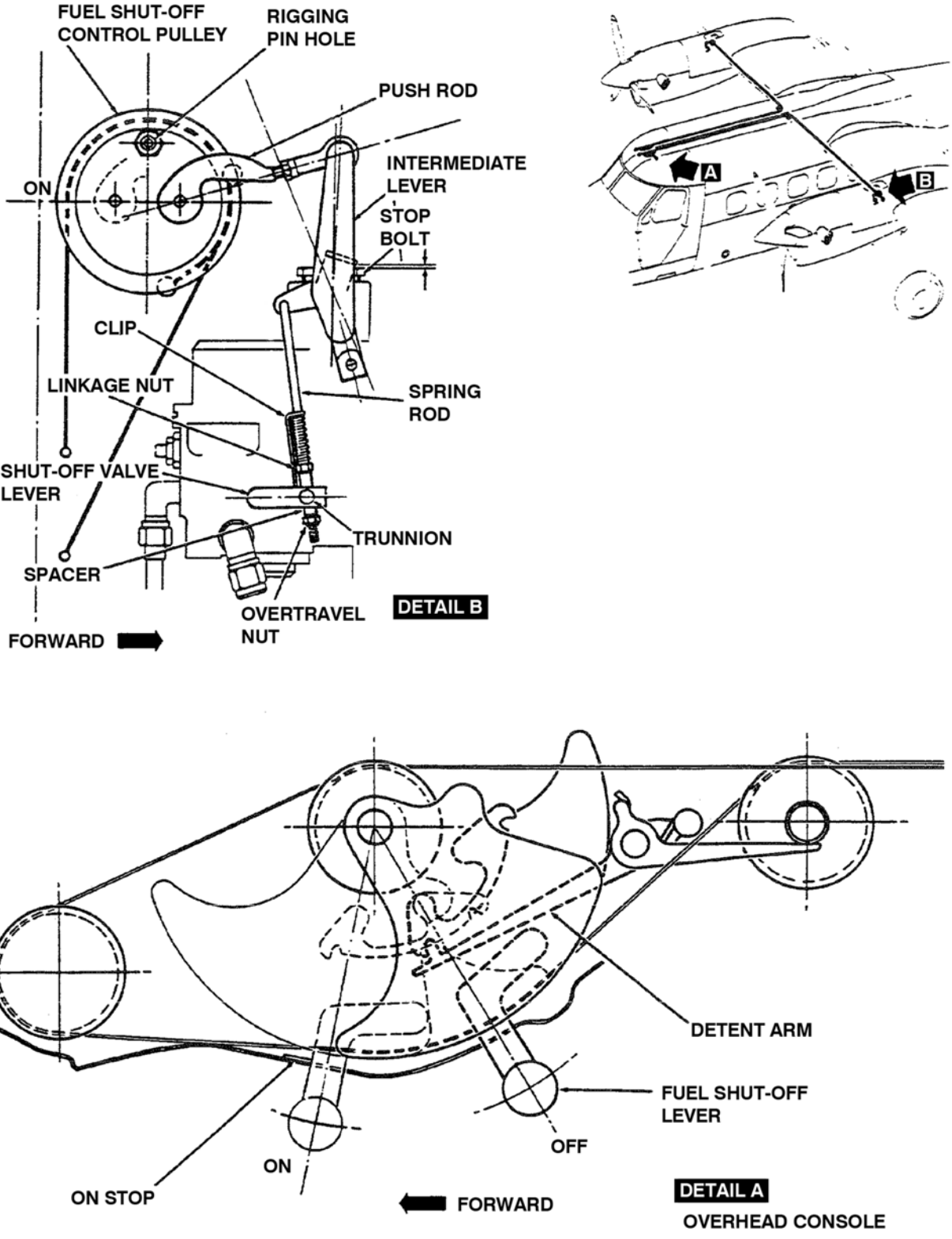
6) Remove rig pin from the nacelle.

7) Move engine fuel condition lever to the ON position.

8) Verify that the FCU intermediate lever is aft of the perpendicular position indicating the FCU shut-off is in the fuel ON position.

9) Ensure fuel shut-off control lever moves freely with no binding.

10) Tighten and safety all control hardware. Carry out function test of the FCU fuel shut-off system.




Rigging Fuel Shut-off Controls
Figure 3

5. Rig Propeller Controls

Refer to Figure 4 for details.

A. Adjust Controls in Airframe

1) Disconnect push-pull control at speed select lever on propeller governor unit.

2) Move propeller lever in flight compartment to minimum rpm position detent (roller against stop face). Apply friction lock.

3) Insert rigging pin SD10545–1 in propeller control pulley.

4) Adjust turnbuckles in wing leading edge to obtain cable tension shown in Figure 2.

5) Ensure propeller lever is not moved from minimum rpm position detent and rigging pin moves freely in rigging pin hole of propeller control pulley.

6) Lock turnbuckles.

B. Adjust Controls – Airframe to Engine Interconnect

1) Install rigging pin SD10545–1 in propeller control pulley bracket in engine nacelle.

2) Ensure that propeller governor unit speed select lever is set at minimum rpm position.

3) Adjust sliding rod end so as to achieve a gap of 0.230 inch between plate and feather stop bolt (refer to Detail A of Figure 4).

Note Attachment bolt to be in mid slot position on speed select lever (refer to Detail A of Figure 4).

4) Remove rigging pin from propeller control pulley.

5) Ensure friction lock knob is loosened to permit control lever to move freely.

6) Move control lever fully forward until plate on speed select lever is hard against maximum rpm stop bolt (refer to Detail A of Figure 4).

7) Adjust forward stop bolt in overhead propeller lever console so as to have a gap of 0.120 to 0.080 inch.

Note Clearance may be checked with setting pin SD12570–3 or feeler gauges.

8) Check that propeller levers align within 0.100 inch at knobs. Relocate attachment bolt in slot of speed select lever, on the Constant Speed Unit (CSU) as required.

9) With propeller lever forward apply light pressure on lever. Check gap of forward stop bolt and if necessary readjust to achieve 0.120 to 0.080 inch gap.

10) Retard propeller lever until the speed select lever plate contacts the feather stop bolt on governor.

11) With propeller lever fully aft apply light pressure on lever. Check gap of aft stop bolt and if necessary readjust to achieve 0.120 to 0.080 inch gap.

12) Move propeller lever through full range of travel and check for complete freedom of movement.

13) Safety all attachments and stop bolts as required.

14) If a new cable has been installed, adjust the length as follows:

a) Install rigging pin SD10545–1 in propeller control pulley.

b) Ensure that propeller governor unit is at minimum rpm position.

c) Adjust sliding rod end so as to achieve a gap of 0.230 inch between plate and feather stop bolt (refer to Detail A of Figure 4).

15) If the new cable is too long, trim the cable as follows:

**CAUTION** **THE FOLLOWING PROCEDURE FOR TRIMMING CABLE IS FOR P/N C6CE1351–31 ONLY.**

a) Cut the cable with an abrasive cutoff wheel.

Note The cable wires are stress relieved and will not unravel when cut.

b) Grind or file the cable to shape a rounded end. Grind or file the cable end in the direction of the cable winding.

**CAUTION** **DO NOT USE A SOLVENT ON THE CABLE ASSEMBLIES. THE SOLVENT WILL DESTROY THE INTERNAL LUBRICANT OF THE CABLE AND WILL MAKE THE CABLE UNFIT FOR USE.**

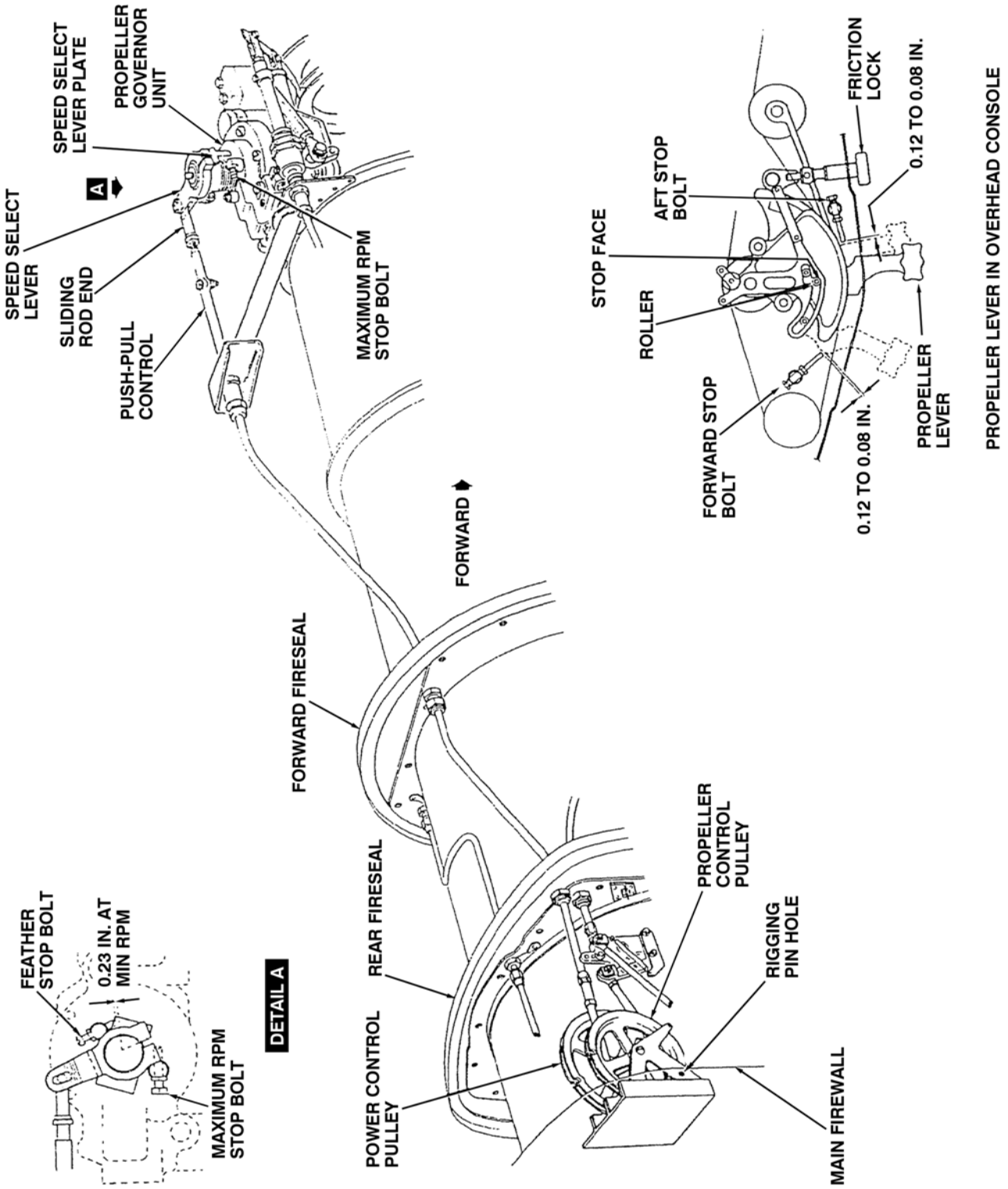
c) Clean the end of the cable by brushing at right angles to the centerline of the cable with a hard fiber, plastic or brass bristle brush. Do not use a steel brush.

d) Using dry, oil free compressed air at no more than 25 psi, blow the loose dirt particles from the cable.

e) Wipe the cable clean with a dry lint free cloth.

Note It is not necessary to apply any protection to the cut cable end to prevent corrosion, fraying and unwinding.

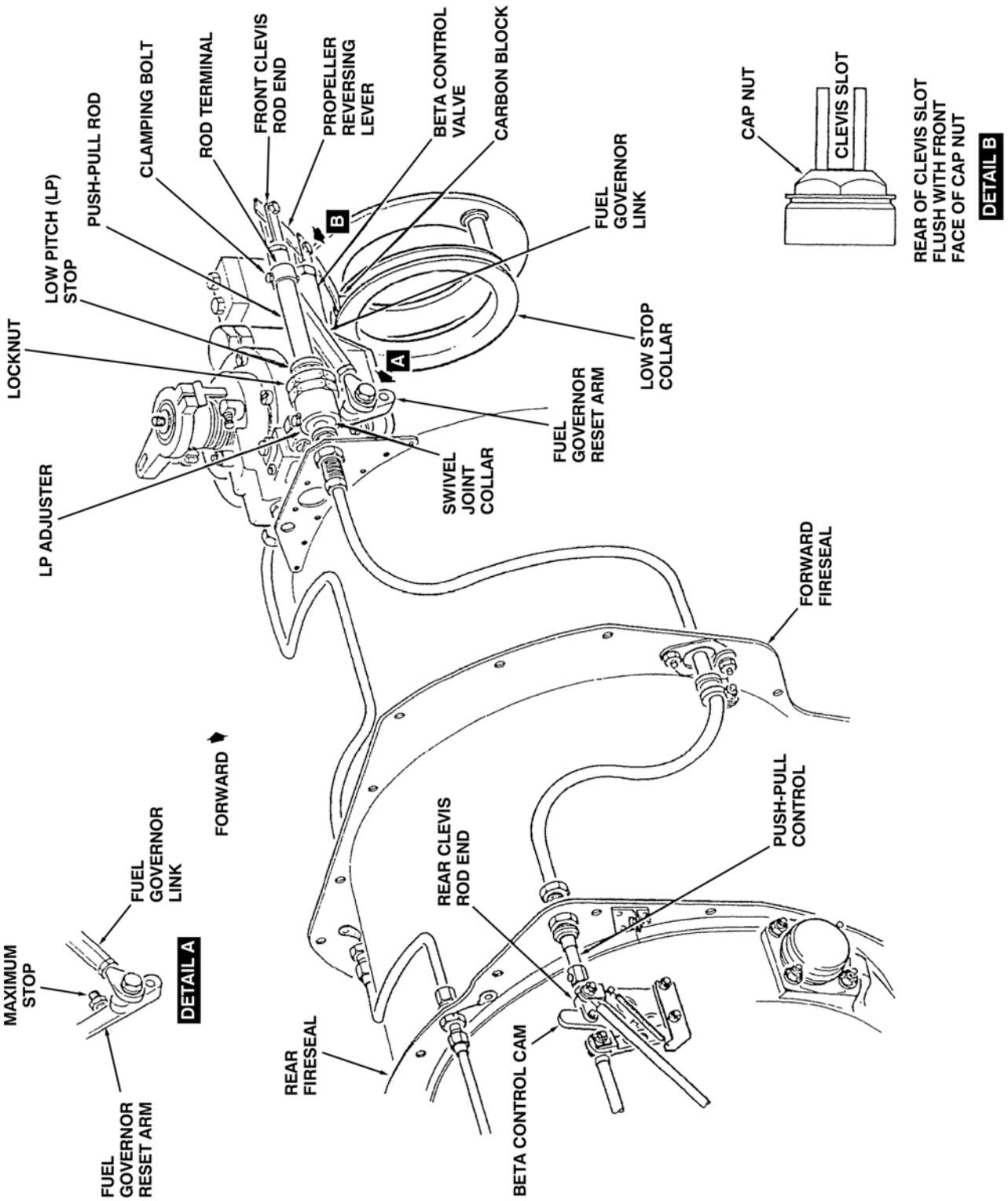
f) Do a visual inspection on the cable for serviceability.




Rigging Propeller Control System
Figure 4



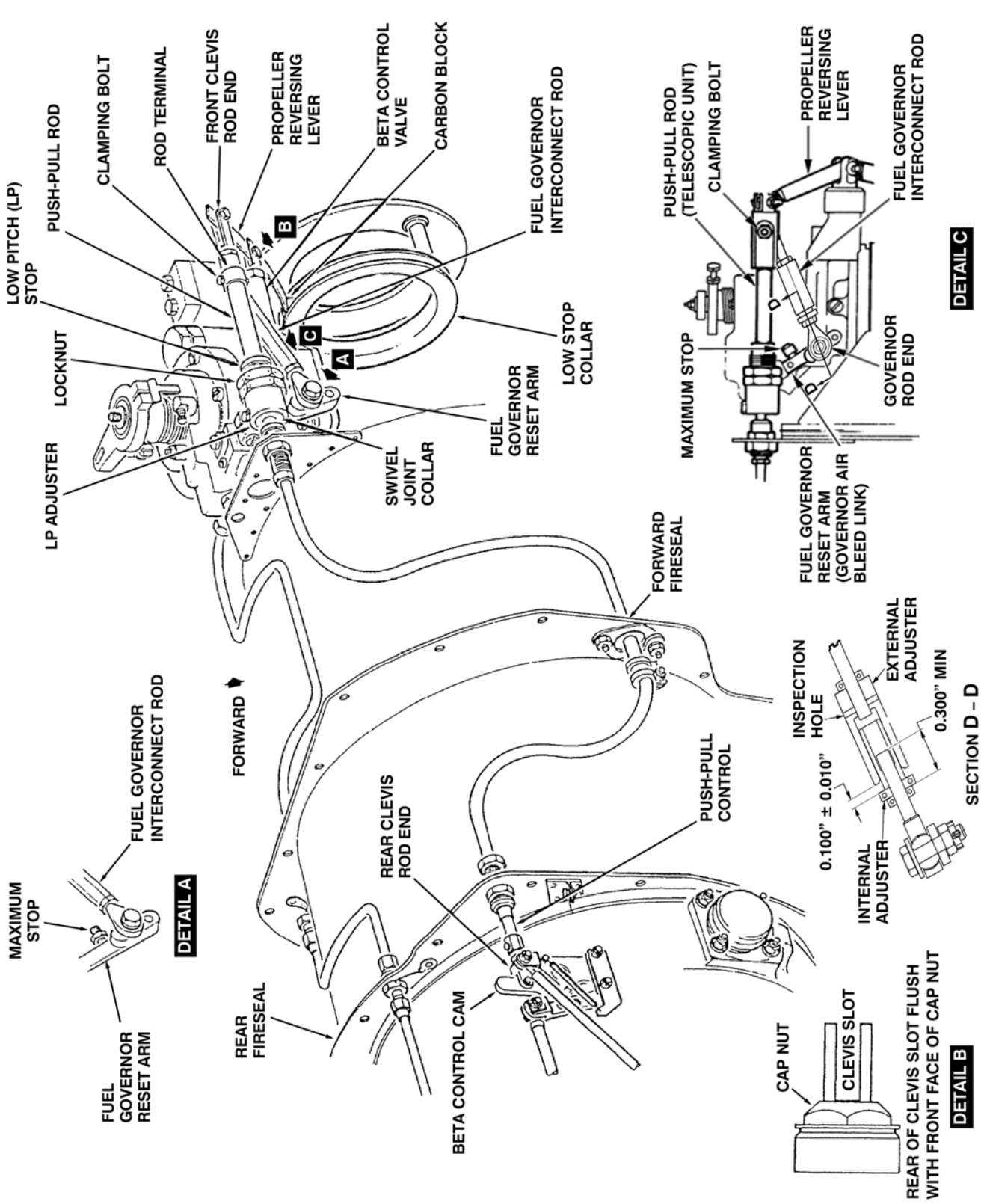
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Rigging Beta Control System (Pre Mod 6/2476)
Figure 5



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Rigging Beta Control System (Post Mod 6/2476)
Figure 6



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6. Rig Cable Slides

Refer to Figure 7 for details.

A. Adjust Cable Slides (Autofeather)

1) Ensure propeller levers are at full INCREASE and with teleflex push-pull control (refer to Figure 5 (Pre Mod 6/2476) or Figure 6 (Post Mod 6/2476)) disconnected at beta control cam, set both power levers to MAX REVERSE. Apply friction lock.

2) Remove upholstery from cabin roof in vicinity of stations 147 to 157.

3) Adjust position of cable slides, if necessary, by loosening bolts securing slides to cable. Move slides so that front chamfers of cable slides are 2.6 inches from rear end of slide tubes as shown in Figure 7.

4) Secure glides by tightening bolts.

5) Release power lever friction lock, move power levers through full range of travel and check for complete freedom of movement.

6) Connect push-pull control to beta control cam.

B. Adjust Cable Slides (Propeller Reset Caution)

1) Move right power lever to IDLE position and apply friction lock.

2) Move left propeller lever to FEATHER position and apply friction lock.

3) Remove upholstery from cabin roof in vicinity of stations 140 to 148.

4) Adjust position of cable slides, if necessary, by loosening bolts securing slides to cables.

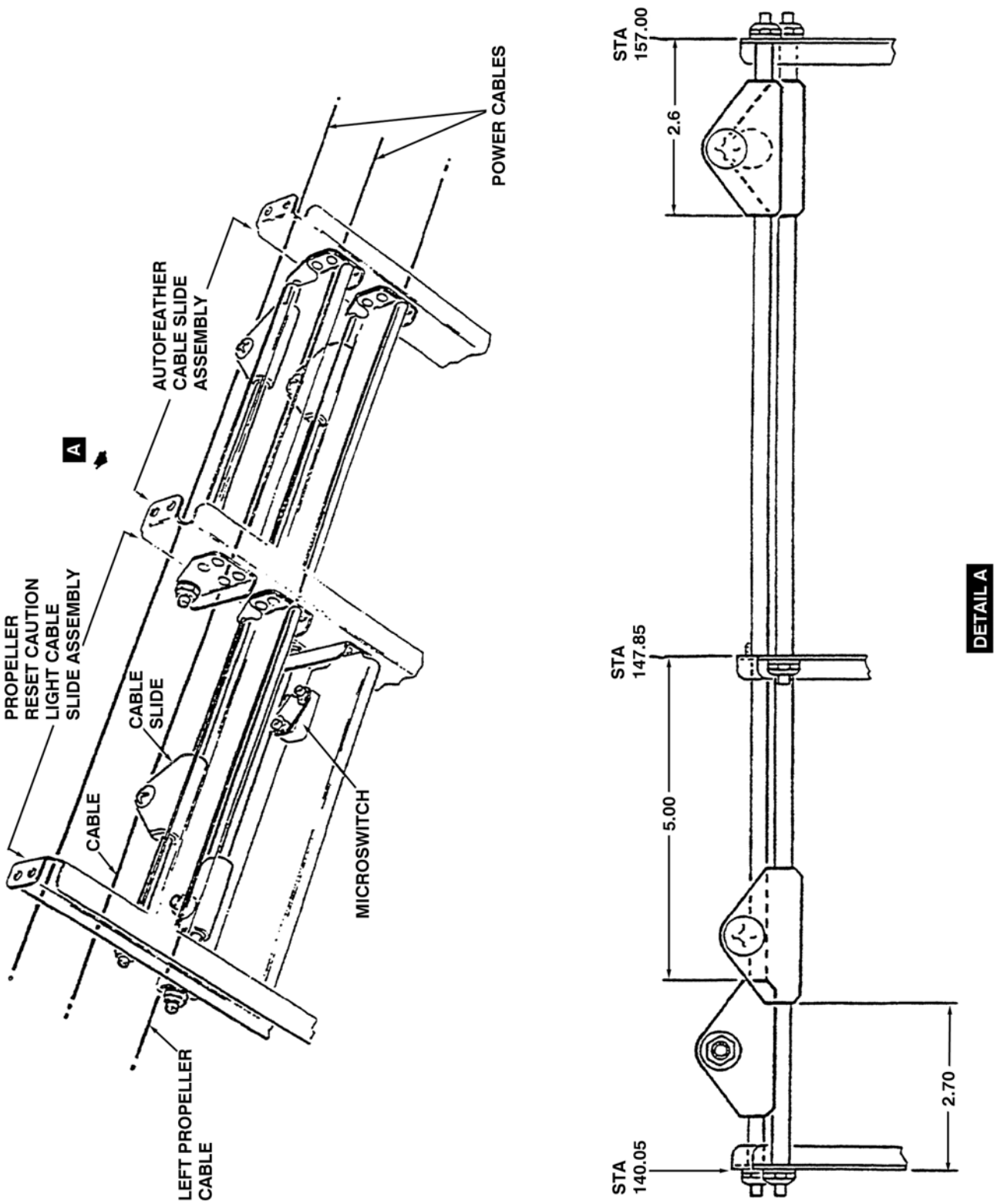
5) Move propeller cable slide (right side) so that rear end of slide is 5.00 inches from rear end of slide tube as shown in Figure 7.

6) Secure propeller cable slide by tightening bolt.

7) Move cable slide (left side) so that front end of slide is 2.7 inches from front end of slide tube, as shown in Figure 7.

8) Secure power cable slide by tightening bolt.

9) Release power and propeller lever friction locks. Check freedom of controls by moving propeller levers through full range of travel and power levers through forward range.




Rigging of Cable Slides
Figure 7



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